

Thomas Davis

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Full stack developer with 15+ years building products from the ground up at startups and open source. Recently focused on generative AI — building LLM-powered applications with Vercel AI SDK, designing MCP tool ecosystems, and creating AI agent infrastructure. Product-minded engineer who ships fast and prioritizes user feedback.

Experience

Product Engineer (AI Focus)

May 2020 — Present

Misc Companies, Melbourne

Working at early-stage startups building generative AI products. Building full stack applications with Vercel AI SDK, designing LLM pipelines with structured outputs, implementing vector search and RAG systems, and creating AI agent tooling including MCP servers and Claude Code skills. Managing cloud infrastructure on GCP with Kubernetes for production workloads.

- Vercel AI SDK / OpenAI / Anthropic
- Structured outputs, embeddings, vector search
- MCP servers and AI agent infrastructure
- GCP / Kubernetes / Cloud Infrastructure
- React / Next.js / TypeScript
- Node.js / PostgreSQL / Supabase

Senior Javascript Developer

May 2020 — May 2021

Tokenized, Melbourne

Tokenized is a Bitcoin wallet for issuing, managing and trading digital tokens. I built out the front end which was packaged as an electron app. It was a difficult frontend to build because we store the users keys locally and used them to sign transactions and contracts.

- React
- Redux
- SCSS
- Product

Senior Javascript Developer

Mar 2018 — Jan 2020

Blockbid, Melbourne

Blockbid is an Australian crypto currency exchange. I started off on the frontend but eventually became lead tech and worked on every moving piece of the exchange. I really enjoyed working with liquidity providers and connecting their platforms to help us achieve liquid markets.

- React, Apollo, Styled Components
- Node.js / Rails
- Docker / Heroku / GCP
- Used Figma for design and UX work
- Optimizing markets with tens of millions of rows using SQL.

Developer

Jan 2016 — Jan 2018

Listium

Built a very large and complex React / Redux application. It works on all platforms and has IOS/Android builds due to it being a PWA. (wrapped it in React Native though only implementing a WebView)

- Worked with Postgres, Redis and Dynamodb for storage.
- Hosted on a mixture of Heroku Apps and EC2 servers.
- Caching by Fastly and Cloudflare
- Hybrid app supported on all platforms

Developer

Apr 2014 — Jan 2016

Electronic Frontier Foundation

Hired to take EFF's campaigning to the next level by building an action centre in Ruby on Rails. The action centre is built around some large open source tools that lower the barrier to entry when contacting congress.

- Developed new tools for contacting congress
- Brainstormed campaign ideas to raise maximum awareness about issues
- Lots of social networking integration

CTO

Mar 2013 — Jan 2015

Earbits

Started off as a front end developer but took on the role of CTO in early 2013. The application frontend is built with Javascript and organized as a single page application that talks to a collection of Rails web servers which are connected to MongoDB.

- Managed a small team of developers and designers
- Built the entire frontend application with Backbone.js
- Transferred all of the infrastructure from Heroku to AWS

Founder

Apr 2014 — Present

JSON Resume

JSON Resume is a community driven open source initiative to create a JSON based standard for resumes. There is no reason why there can't be a common standard for writing a resume that can be extended with an ecosystem of open source tools.

- This resume is built with JSON Resume
- Over 5000 stars on Github
- Community developed themes
- Tens of thousands of users

Developer

Jun 2013 — Jan 2016

Taskforce.is

Worked on many politically charged campaigns against mass surveillance. Not only technically challenging work but also a lot of networking and getting my hands dirty with politics. Our biggest project was "TheDayWeFightBack"..

- Generated 37,000,000 banner views
- 100, 000 phone calls to congress
- 500, 000 emails
- 250, 000 signatures

Co-Founder

Jan 2013 — Present

Drone Hire

An international directory of civilian drone / UAV operators for hire. Services include aerial photography, aerial video, mapping, surveying, precision agriculture, real estate photography, remote inspections and infrared imaging. Our plan is to be the place to go when looking for UAV/Drone services. The website is built in Backbone.js and API is built with Node.js and Postgres.

- The site and blog combined have managed to receive over 200,000 visitors in 2014.

Founder

Jan 2011 — Jan 2014

BackboneTutorials.com

I write tutorials and blog post regarding the popular Javascript framework Backbone.js. The tutorials cover a range of topics regarding front end development aimed at beginners, experts and anyone in between.

- Over two million unique visitors a year
- 25,000+ ebook downloads
- 300,000+ Youtube views

Front-end Developer

Jan 2011 — Jun 2012

Ephox

Ephox is a worldwide company who is heavily involved with the development of TinyMce and enterprise editors. My primary role included building front-end widgets and applications. Worked on a major product using Backbone.js as a base. Heavily involved in UI/UX design and wire-framing. Also spend a lot of time designing API specifications and documentation.

Volunteering

Contributor

Jan 2016 — Present

Open Source Project

- Fixed critical bugs and added new features.
- Mentored new contributors.

Projects

- **JSON Resume:** 5000+ GitHub stars, tens of thousands of users. AI job matching with vector embeddings and semantic search. Built with Next.js, Supabase, Vercel AI SDK. Community developed themes ecosystem.
- **Cdnjs:** Millions of sites use the CDN in production. Larger market share than Yahoo's, Microsoft's and Google's javascript content distribution networks. We serve hundreds of billions request a month. Contains over 3000 popular Javascript libraries.
- **Omega:** 80+ AI tools including code execution in 42+ languages. Self-evolution capability — analyzes behavior and proposes improvements via GitHub issues. Dual database architecture (PostgreSQL + MongoDB). AI-powered decision making for natural conversation participation.
- **Alpha:** Custom tensor backend with tape-based reverse-mode autodiff. Hand-written Vulkan GPU backend with TS-generated SPIR-V kernels. Full training pipeline: BPE tokenizers, checkpointing, eval, sampling. OpenAI-compatible chat API, CLI, web dashboard, and TUI.
- **TPMJS:** Auto-discovers npm packages with tpmjs keyword. Sandboxed tool execution via Deno runtime. MCP aggregator for connecting tool collections to Claude, Cursor, etc.. Agent platform for building, sharing, and forking AI agents.
- **Blocks:** Domain specification via blocks.yml for entities, signals, measures, and rules. Multi-layer validation: Schema, Shape, and AI-powered Domain checks. Semantic feedback loop for AI coding tools. Drift detection between code and specification.
- **MobTranslate:** Implemented multilingual dictionary with advanced search and pagination.. Added real-time translation API using AI techniques.. Optimized performance with server-side rendering (Next.js).. Structured codebase as a monorepo using Turborepo..

Education

University of Oklahoma

Jun 2011 — Jan 2014

Bachelor - Information Technology, GPA: 4.0

Publications

Video compression for 3d media

Oct 2014

Awards

Defender of the Internet, Fight For The Future

Oct 2014

Skills and Tech

- **AI & LLMs:** Vercel AI SDK, OpenAI API, Anthropic API, Google Gemini, Structured Outputs, Embeddings / Vector Search, RAG Pipelines, AI Agents, Model Context Protocol (MCP), Prompt Engineering, LLM Training (from scratch), Autograd / Tensor Operations.
- **Frontend:** React, Next.js, TypeScript, JavaScript, HTML / JSX, Tailwind CSS, Styled Components, Redux, Apollo / GraphQL, Electron, Three.js, Backbone.js.
- **Backend:** Node.js, Go, Python, Ruby on Rails, PostgreSQL, MongoDB, Redis, Supabase, REST APIs, WebSockets, Deno.
- **Cloud & DevOps:** Google Cloud Platform (GCP), Kubernetes, AWS, Docker, Vercel, Railway, Heroku, GitHub Actions, CI/CD Pipelines, Serverless, Cloudflare.

- **Architecture & Tooling:** Monorepos (Turborepo), System Design, API Design, Microservices, Real-time Systems (Pusher / Yjs), CLI Tools, npm Package Publishing, Open Source Maintenance.
- **Low-Level & GPU:** C, Vulkan, SPIR-V Shader Generation, GPU Compute, Mixed Precision (FP16).

Interests

- **Gardening:** Lazy Gardening
- **Music:** Guitar, Singing, Dancing
- **Books:** Reading, Writing, History
- **Open Source:** All of it